

Theorem. For a fixed vector a , if a sequence of random vectors Y_n converges in distribution to a random vector Y , then $a \cdot Y_n$ converges in distribution to $a \cdot Y$, and that if $a \cdot Y_n$ converges in distribution to $a \cdot Y$ for all vectors a , then Y_n converges in distribution to Y . The first part is because a linear combination of the components of the Y_n converges in distribution to the linear combination of the components of Y since Y_n converges to Y . The other direction is the hard part: Note that it is not as obvious as it seems: Just because each component converges to the right thing does not mean the whole thing does, since the components are not independent in general.

In fact, by having a distribution Y and setting the sequence to be $Y, Y, Y, \dots$ the above theorem actually says that every probability distribution in higher dimensional space is uniquely determined by its 1D projections, or that scanning something in every direction determines its structure, which is really interesting. (This is for distributions which are part discrete part continuous – which is more than what we need – since our characteristic function theory only works for those types of distributions)

Levels recommended for proof: 6 (A good understanding of all the hard stuff in level 6 stats and technical results is expected)

Lemma 1. Convergence in distribution implies pointwise convergence of characteristic function

Proof. See level 6.3

□

Definition. A random vector is defined by a probability distribution function in $\mathbb{R}^k$, i.e. sets of k real numbers. We define the cf of a random vector Y to be a function that takes in a vector t and outputs $E(e^{i(t \cdot Y)})$.

Lemma 2. Let Z_n, Z be random vectors such that $Z_n \rightarrow Z$ in distribution (meaning cdfs converge at all continuity points of Z 's cdf). Let t be any vector of equal dimension, then $t \cdot Z_n \rightarrow t \cdot Z$ in distribution.

Proof. Unfinished

□

Lemma 3. Lemma 1 extends to vectors.

Proof. Unfinished

□

Lemma 4. Random vectors have their characteristic function determine the distribution.

Proof. If the space is d dimensional, we define

$$I_\varepsilon := \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \phi_y(t) e^{-\varepsilon|t|^2} \prod_{j=1}^d \frac{e^{-ia_j t_j} - e^{-ib_j t_j}}{it_j} dt$$

Where that big scary looking thing just means product.

$$= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{i(t \cdot y)} \prod_{j=1}^d \frac{e^{-ia_j t_j} - e^{-ib_j t_j}}{it_j} d\mu e^{-\varepsilon|t|^2} dt$$

Note: Each $\frac{e^{-ia_j t_j} - e^{-ib_j t_j}}{it_j}$ is bounded for the same reasons as before (ie in level 6 stats), so we have the product of bounded things times a thing which integrates to 1 in the inner dy integral, so that's bounded. Now in total we have a bounded thing times the integral of $e^{-\varepsilon|t|^2}$, which is finite, so we have the conditions to swap the integrals around.

Since $e^{-\varepsilon|t|^2}$ is just the product of $e^{-\varepsilon(t_j)^2}$, we can simplify I_ε as follows:

$$\begin{aligned} &= \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} e^{i(t,y)} \prod_{j=1}^d \frac{e^{-ia_j t_j} - e^{-ib_j t_j}}{it_j} e^{-\varepsilon|t|^2} dt d\mu \\ &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \prod_{j=1}^d \frac{1}{2\pi} e^{i(t_j y_j)} \frac{e^{-ia_j t_j} - e^{-ib_j t_j}}{it_j} e^{-\varepsilon(t_j)^2} dt d\mu \end{aligned}$$

We know what each term looks like from earlier, so we end up with

$$= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} \prod_{j=1}^d \frac{1}{2} \left(\operatorname{erf} \left(\frac{y_j - a_j}{2\sqrt{\varepsilon}} \right) - \operatorname{erf} \left(\frac{y_j - b_j}{2\sqrt{\varepsilon}} \right) \right) dt d\mu$$

Which in the limit vanishes exactly when we are outside the (a,b) high dimensional rectangle and goes to 1 when we are inside it, for the same reasons – Each term goes to 1 or 0 and the product is 1 only when all terms go to 1.

Now at last we have, because of dominated convergence again, the same result for random vectors: the characteristic function determines the distribution.

□

Lemma 5. We have the other direction (Convergence in cf implies convergence in distribution) for random vectors.

Sketch of proof. This proof builds on work from Level 6.3 when we proved a similar statement. The analogous statement to Helly's selection theorem holds in $\mathbb{R}^n$ with the same proof, where in this case 'increasing' means that $x \leq y$ in every component implies $f(x) \leq f(y)$.

Now we will have a sequence of random vector like $(X_1, X_2, \dots, X_n)$ where the distribution of the sequence of each X_j is tight by the same argument as in level 6.3. Therefore, the whole thing is tight by considering the probability of being within a $[-M, M]^n$ box as M gets large. The argument then proceeds as in level 6.3.

The proof of the main theorem is very short now.

Proof. Suppose we have a vector t , and $Z_n := t \cdot Y_n$ and $Z := t \cdot Y$. Suppose also that Z_n converges in distribution to Z . Then we will show that Y_n converges in distribution to Y .

$$\phi_{Y_n}(t) = E \left(e^{i(t \cdot Y_n)} \right) = E \left(e^{i \cdot 1 \cdot (Z_n)} \right) = \phi_{Z_n}(1)$$

Since Z converges in distribution, we have that as n goes to infinity

$$\phi_{Y_n}(t) = \phi_{Z_n}(1) \rightarrow \phi_Z(1) = E(e^{iZ}) = E(e^{i(t \cdot Y)}) = \phi_Y(t)$$

This is true for every fixed vector t , so since we know that convergence in characteristic functions implies convergence in distribution even for vectors, the result follows.

□